Why Does Ionisation Fail in SF_6?

Three-tier kinetics framework + hybrid spatial surrogate

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The Problem

- SF_6: attachment resonance at 0–1 eV + ionisation threshold at 15.8 eV
- Closure cliff at $T_e \approx 3$ eV: below the knee, $k_{\mathrm{att}} \gg k_{\mathrm{iz}}$
- MCC simulations produce **zero net ionisation**
- Four candidate explanations — which one is right?

Hypothesis Elimination

Hypothesis	Test	Result
M1: EEDF shape	Boltzmann vs Maxwellian	Rejected
M2: Non-local transport	Skin-depth scan	Rejected
M3: Wall losses	Elastic reflector	Rejected
M4: Power density	Double peak field	Accepted

Ionisation failure is a power-density limitation.

Three-Tier Solution

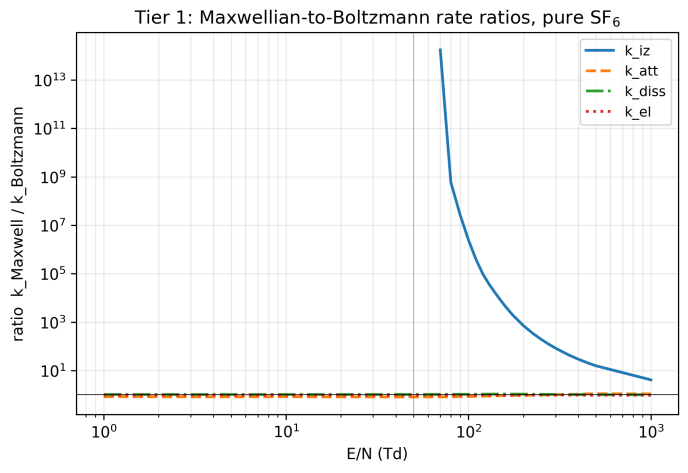
Tier 1 BOLSIG+ Boltzmann solver $\rightarrow$ rate tables, EEDFs

Tier 2 Supervised MLP surrogate (19k params) $\rightarrow$ `get_rates_pinn()` API

Tier 3 Non-local PINNs + MCC cross-check $\rightarrow$ spatial corrections

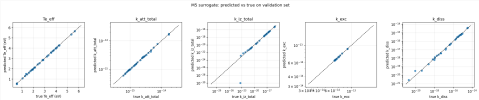
Each tier has a decision gate. Data flows strictly downward.

Tier 1: Rate Ratios



- k_{att} ratio median: **1.11** (barely moves)
- k_{iz} at 1000 Td: **0.24** (Boltzmann reduces ionisation)

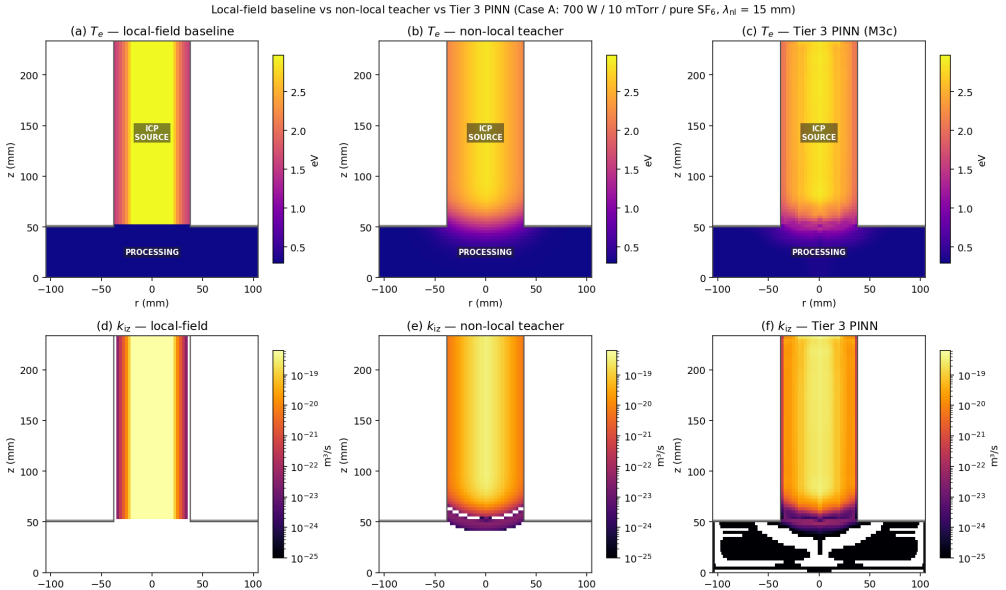
Tier 2: MLP Surrogate



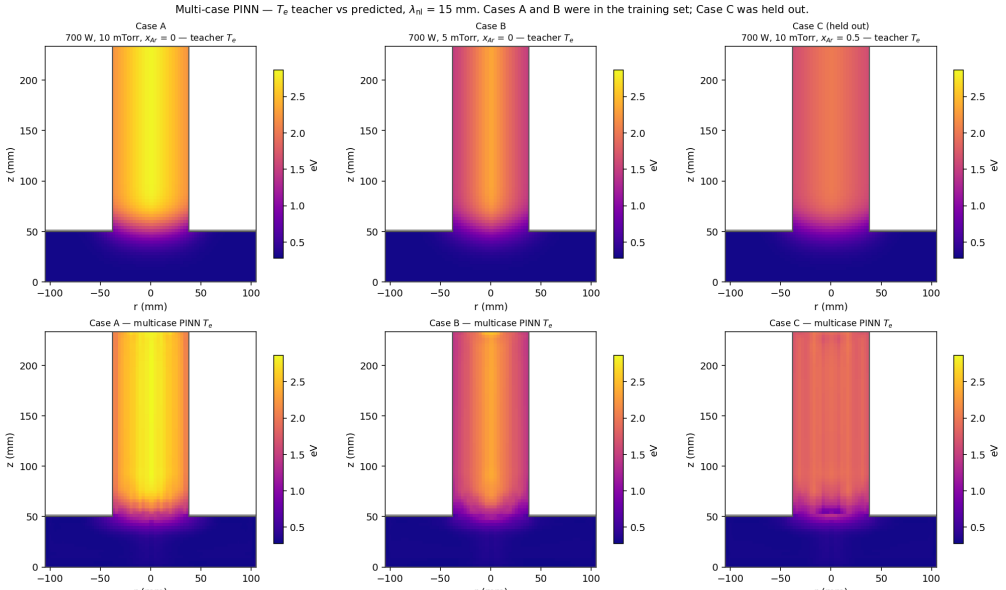
Channel	Median error (%)
T_e	0.44
k_{att}	1.7
k_{iz}	8.4

All channels below 10% gate. Production API delivered.

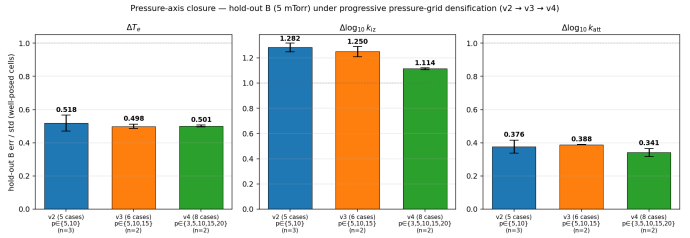
Tier 3: Non-local PINN on TEL Mesh



Tier 3: Multicase Generalisation



Tier 3: Pressure-Axis Closure



Dataset	Hold-out B
v2 (5 cases)	1.282 ± 0.034
v3 (6 cases)	1.250 ± 0.041
v4 (8 cases)	1.114 ± 0.008

Hybrid Spatial Surrogate: Why Not Pure PINN?

Attempted:

- Full PDE PINN solver
- Correct autograd derivatives
- All BCs implemented

Result: Diverged.

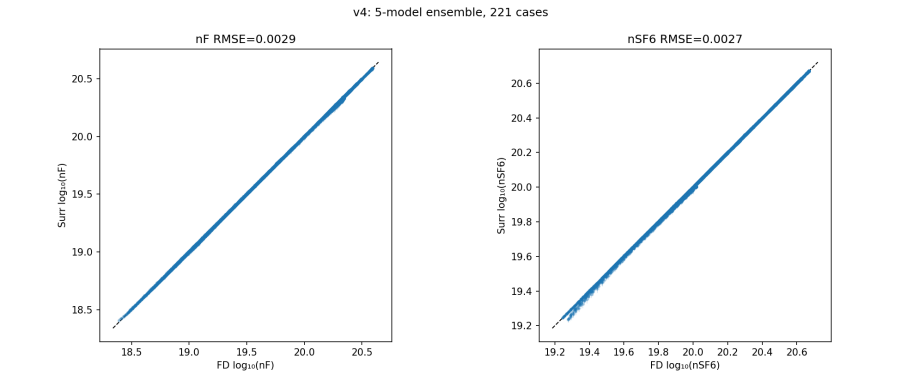
Stiff chemistry + 2nd derivatives

Solution:

- Train on FD solver outputs
- 221-case parametric dataset
- 5-model ensemble + UQ
- Physics regularisation

Result: 0.24% error, 2000× faster

Surrogate v4: Validation



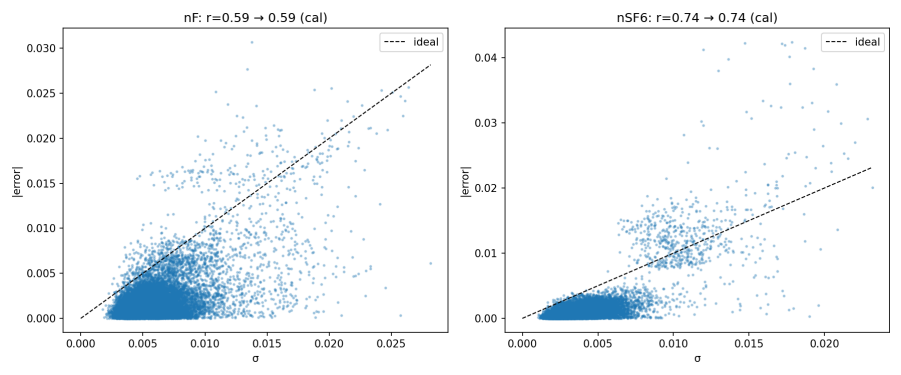
RMSE: $n_{\mathrm{F}} = 0.0029$, $n_{\mathrm{SF}_6} = 0.0027$ ($\log_{10}$ scale)

Surrogate Evolution

	v1	v2	v3	v4
Cases	12	42	88	221
nF RMSE	0.133	0.111	0.023	0.003
nF pct err	19%	14.7%	0.62%	0.24%
Speedup	–	31	2208	1966

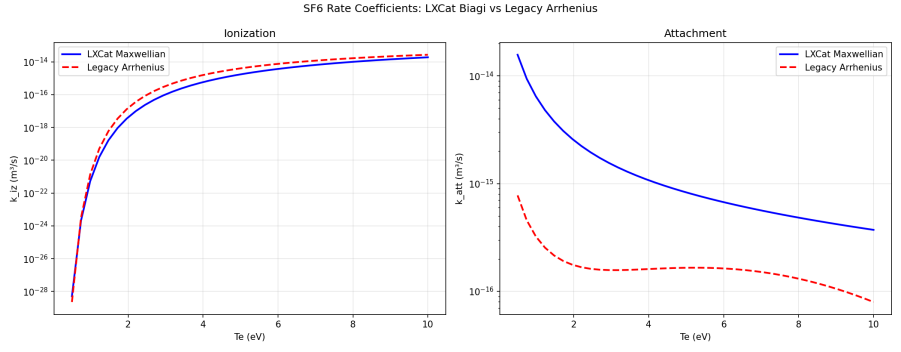
87% RMSE improvement from v3 to v4. Failure corners eliminated.

Uncertainty Calibration



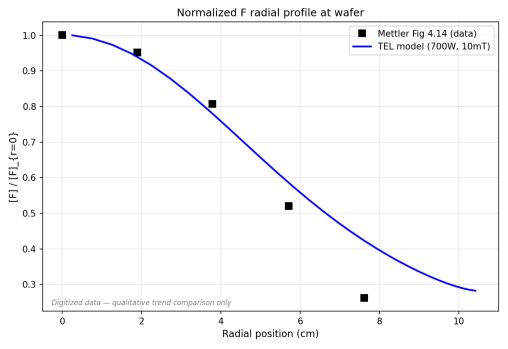
68% coverage achieved after post-hoc calibration.

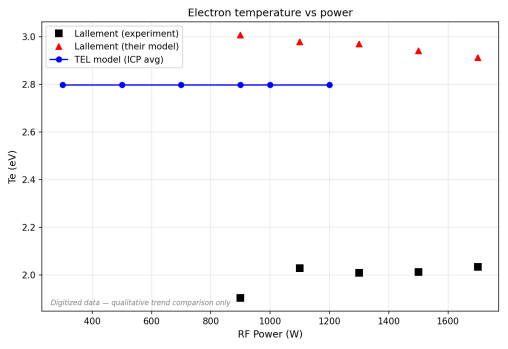
LXCat Cross-Section Upgrade



- LXCat Biagi v10.6: 50 SF_6 + 46 Ar processes parsed
- Ionization: 0.27–0.65× legacy (confirms M1)
- Attachment: 4–20× legacy (SF_6^- resonance)
- Available as alternative rate mode; no existing models broken

Literature Trend Validation





- F radial profile: model drop 72% vs Mettler 74%
- Te: model 2.7 eV vs Lallement 2.0 eV (0D energy balance offset)
- Trends agree; absolute values differ by geometry
- *Digitized data — qualitative comparison only*

Validation Status

Validated:

- Tier 1: BOLSIG+ reference
- Tier 2: $<10\%$ gate met
- Tier 3: rotation + seeds
- Surrogate: 0.003 RMSE

Not validated:

- No experimental data
- PIC-MCC uses synthetic fixture
- Integration uses mock callbacks

Integration and Multi-Species Extension

- Surrogate extended to 3 outputs: n_{F}, $n_{\mathrm{SF_6}}$, T_e
- Integration test: surrogate vs FD at 700W/10mT
- Wafer n_{F} mean error: **0.49%**
- All predictions numerically stable
- Status: **PASS**

Remaining future work (external dependencies only):

1. Real DTPM solver coupling
2. Real PIC-MCC reference data
3. Third $\mathrm{P_{rf}}$ training level for Tier-3 conditioning

One-Slide Takeaway

Ionisation failure in SF_6 is power-limited.

Three hypotheses eliminated. One surrogate delivered.

Non-local correction quantified and bounded.

Hybrid spatial surrogate: 0.24% error, 2000× speedup.

Pure-PINN PDE solver: honest negative result.